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(54) **WIRELESSLY POWERED DECORATIVE
SYSTEMS AND DEVICES**

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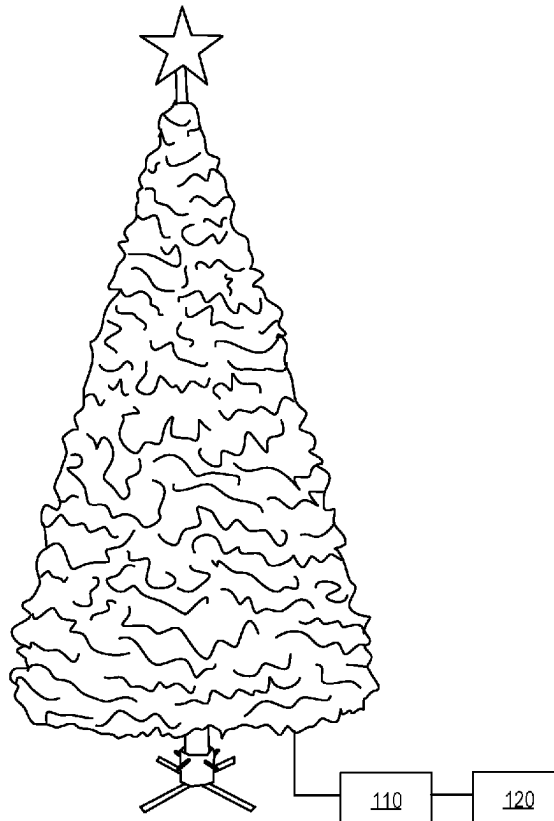
(2013.01); **F21Y 2115/10** (2016.08)

(57)

ABSTRACT

The present disclosure provides a wirelessly powered decorative system. The wirelessly powered decorative system can include a wreath and a wirelessly powered decorative device in communication with the wreath. The wirelessly powered decorative device can include a receiver and a load. The wirelessly powered decorative system further can include a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device. The load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.

100



100

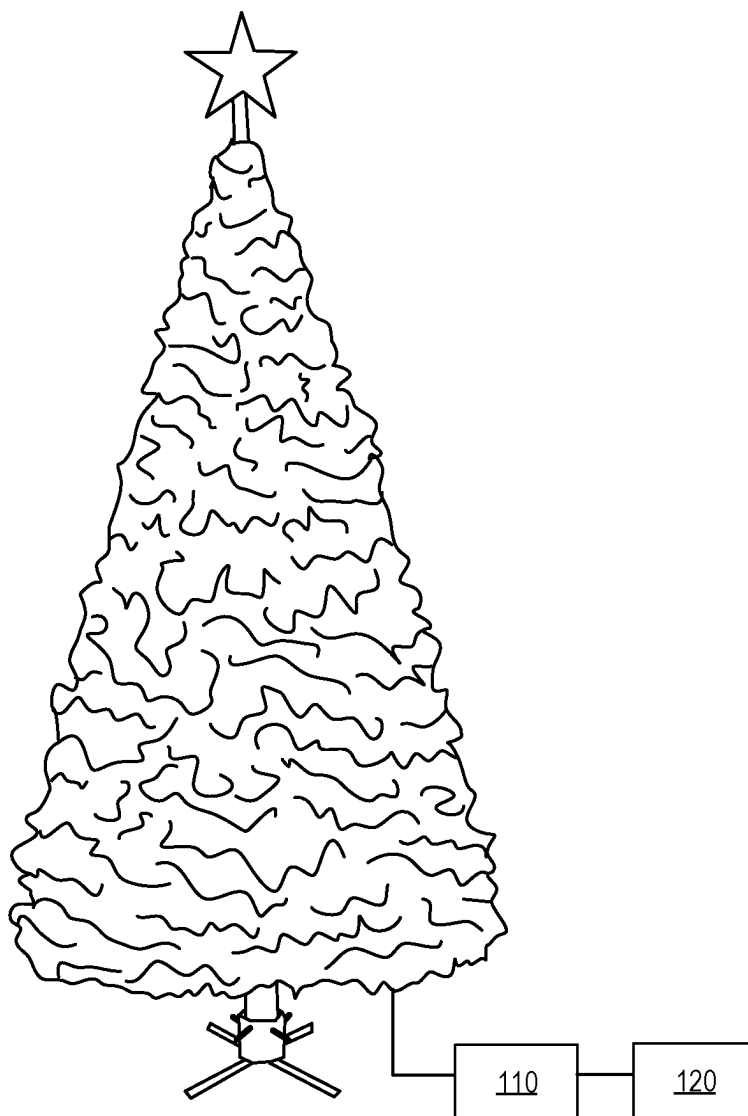


FIG. 1

200

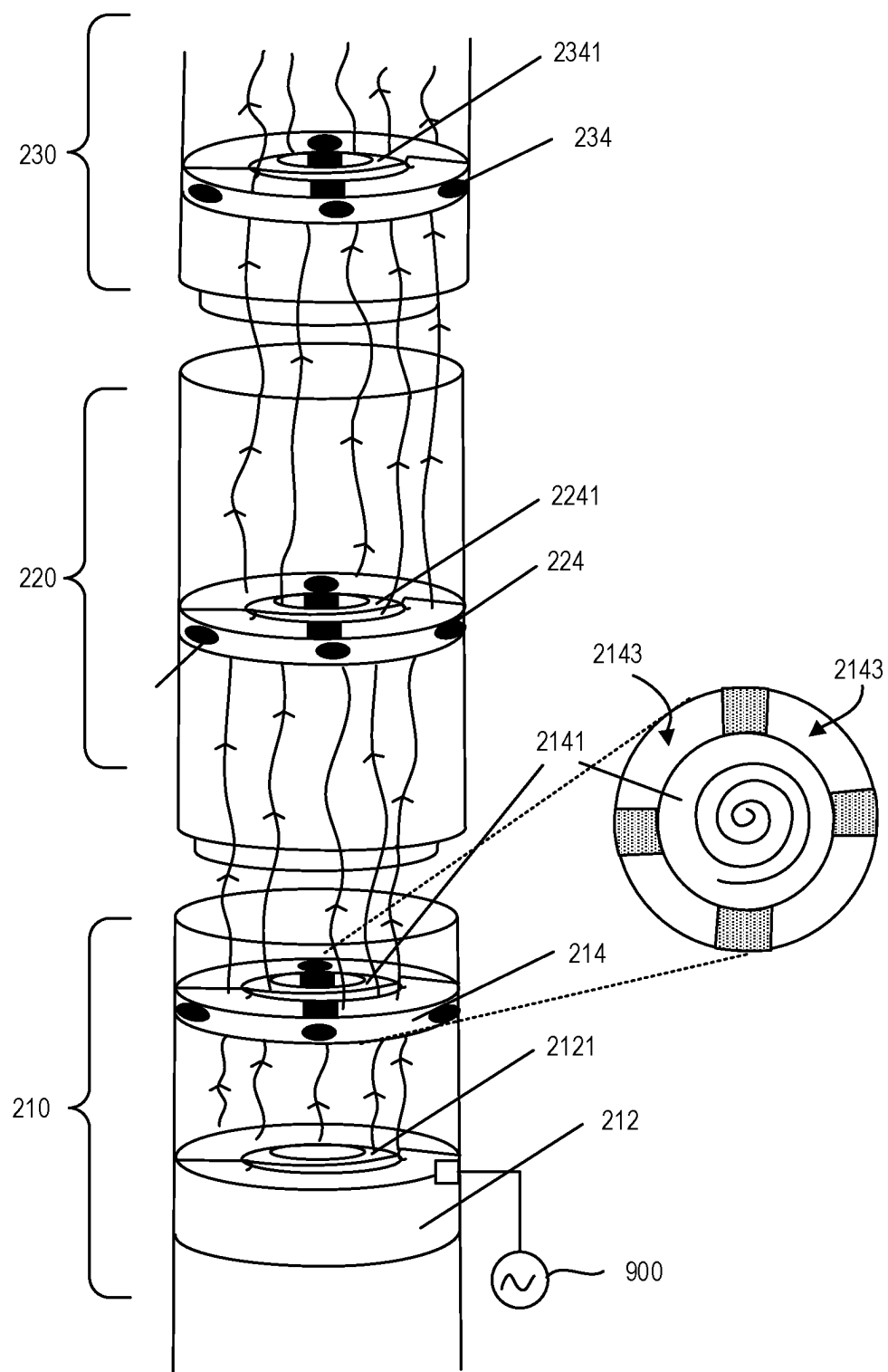


FIG. 2A

100

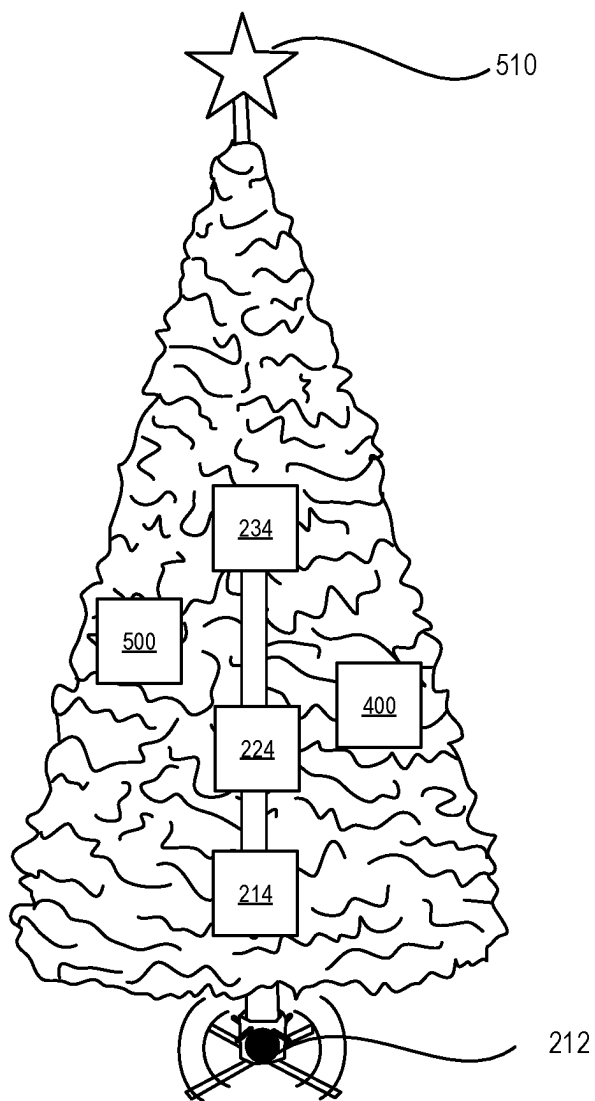


FIG. 2B

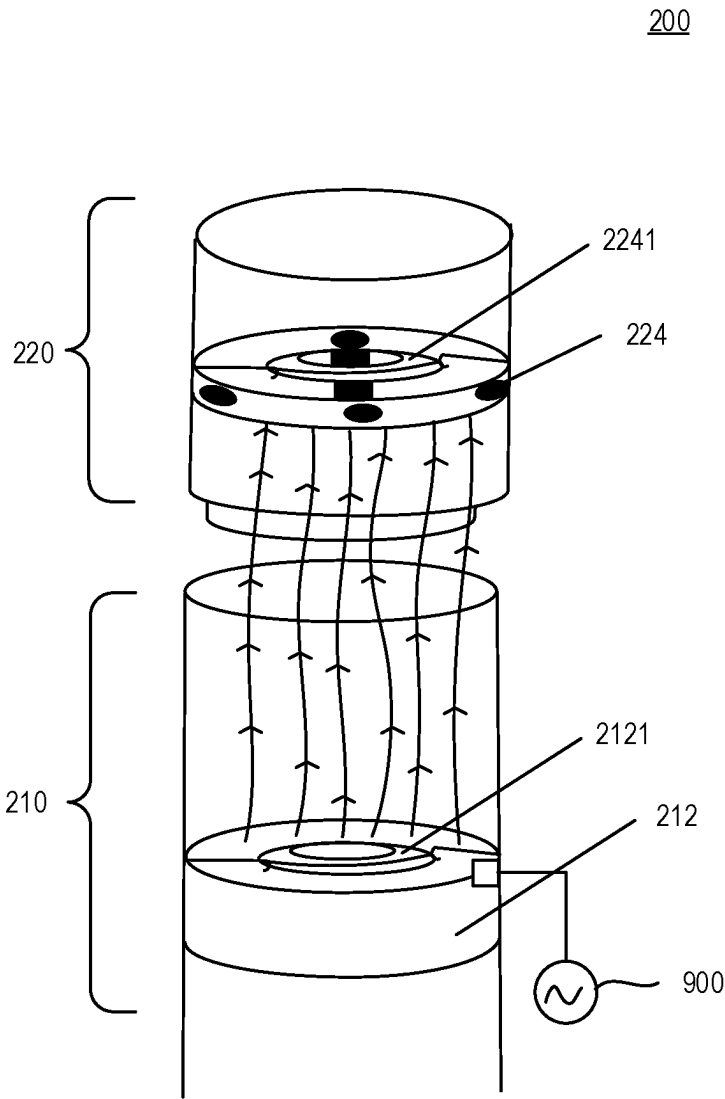


FIG. 2C

200

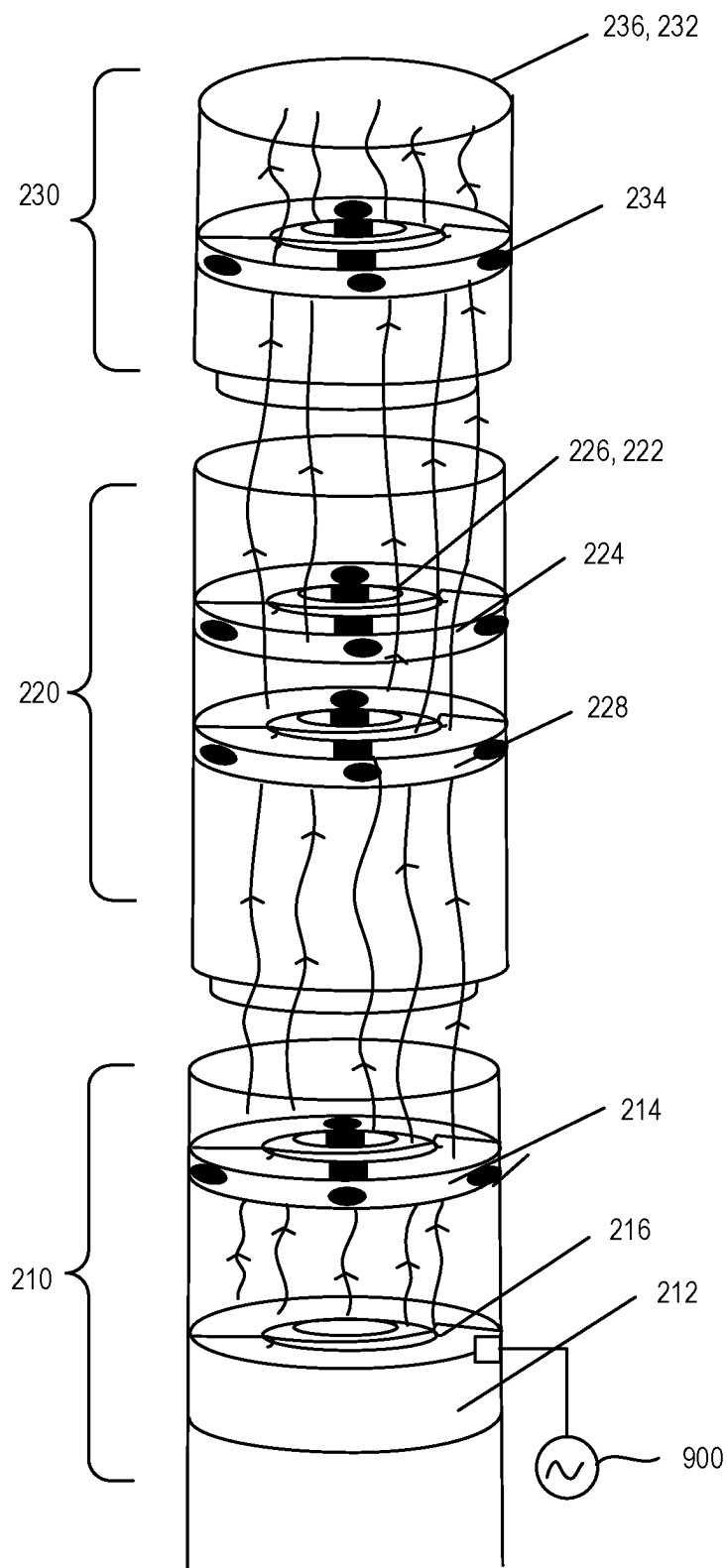


FIG. 3A

200

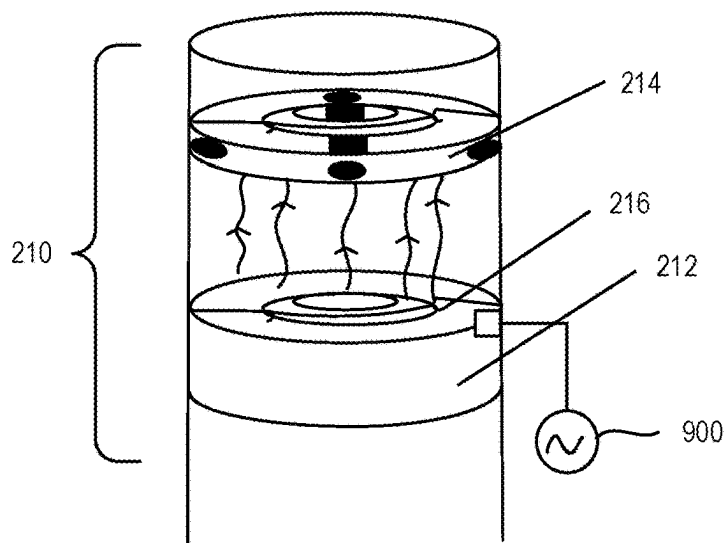


FIG. 3B

200

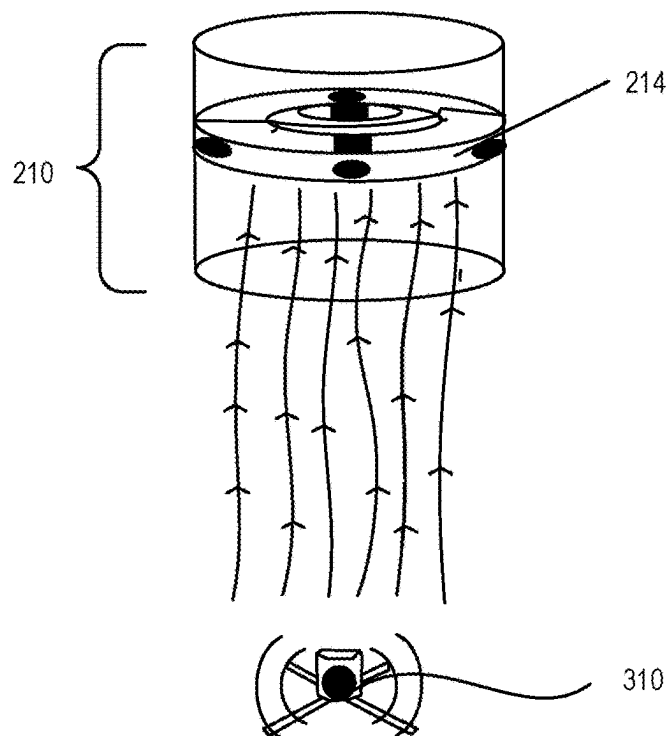


FIG. 3C

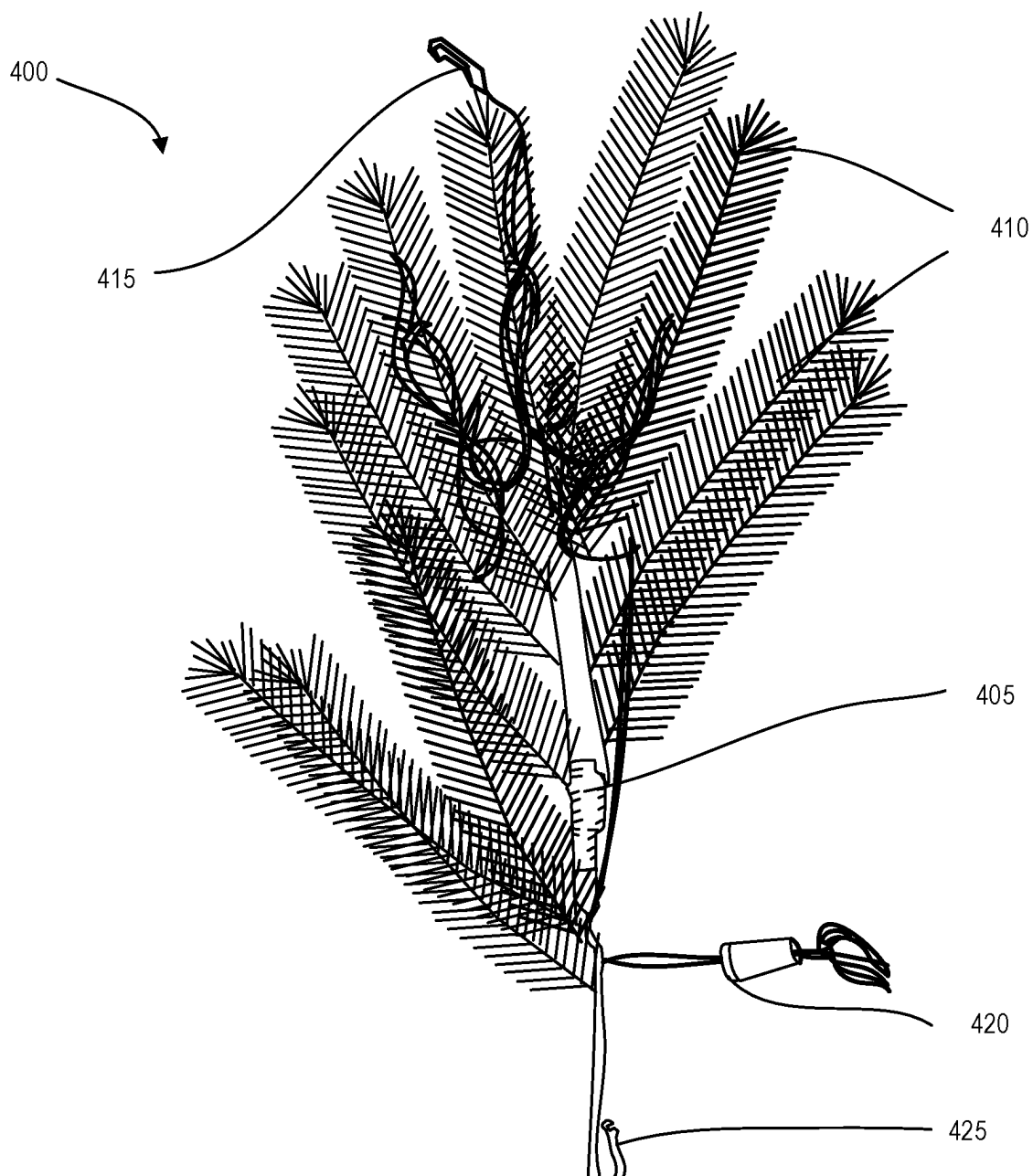
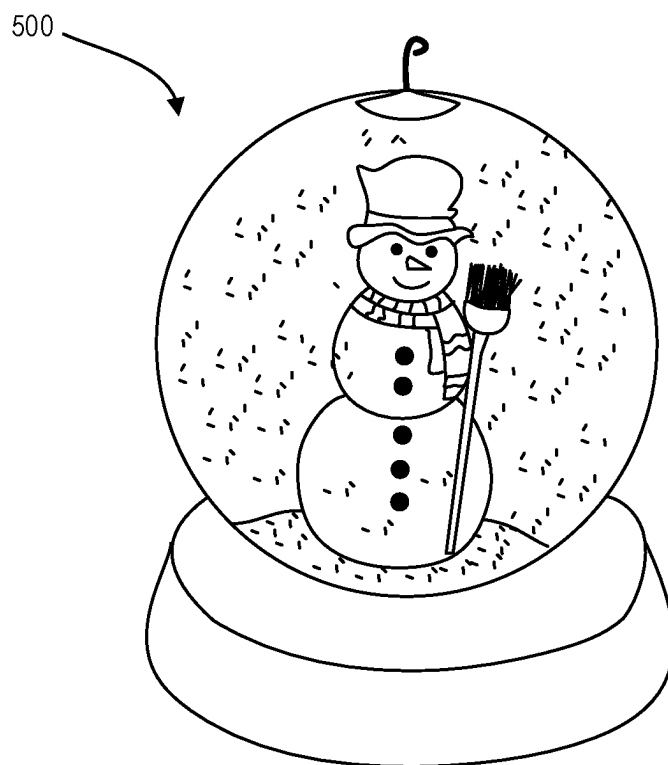
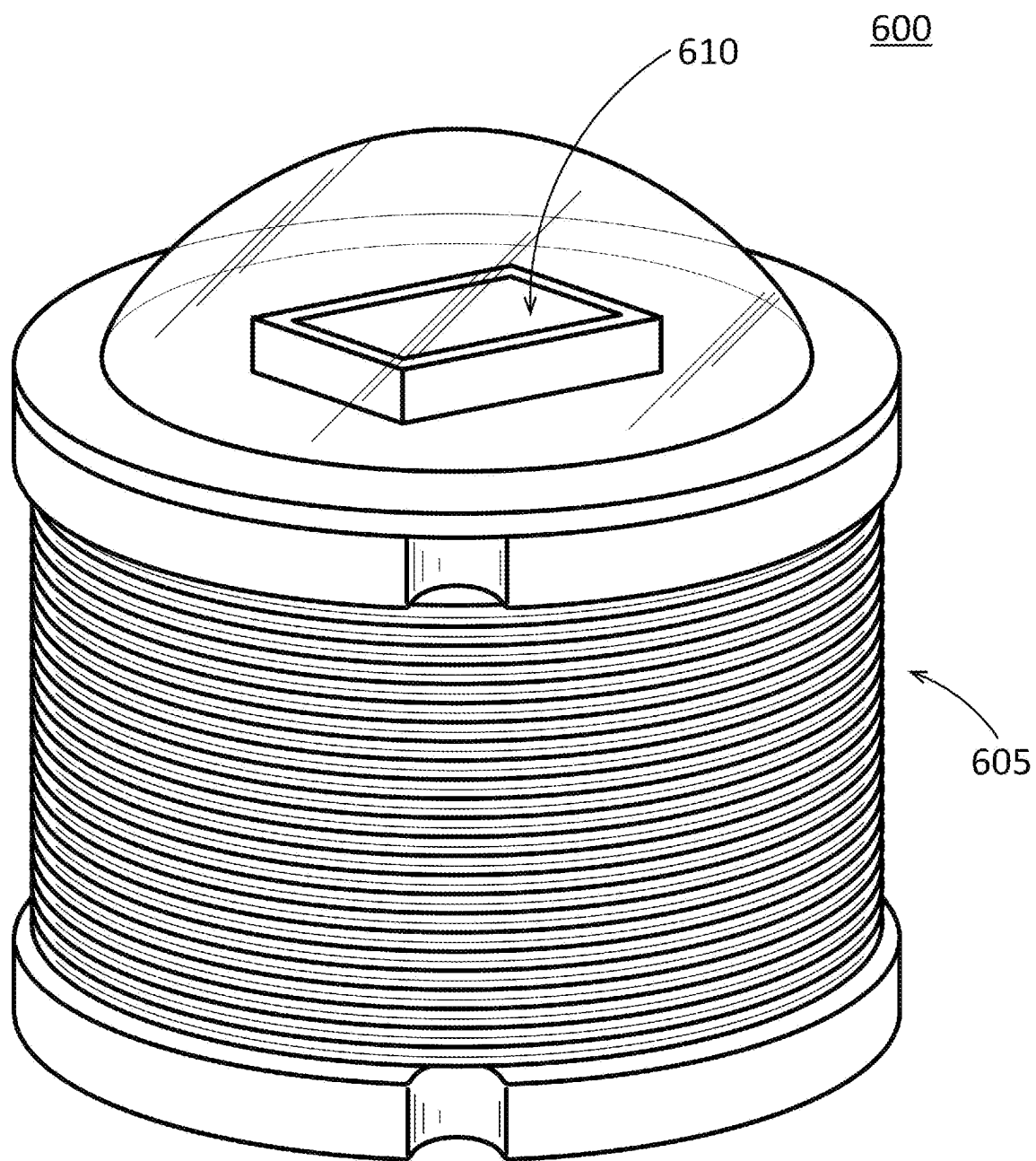


FIG. 4

**FIG. 5**

**FIG. 6**

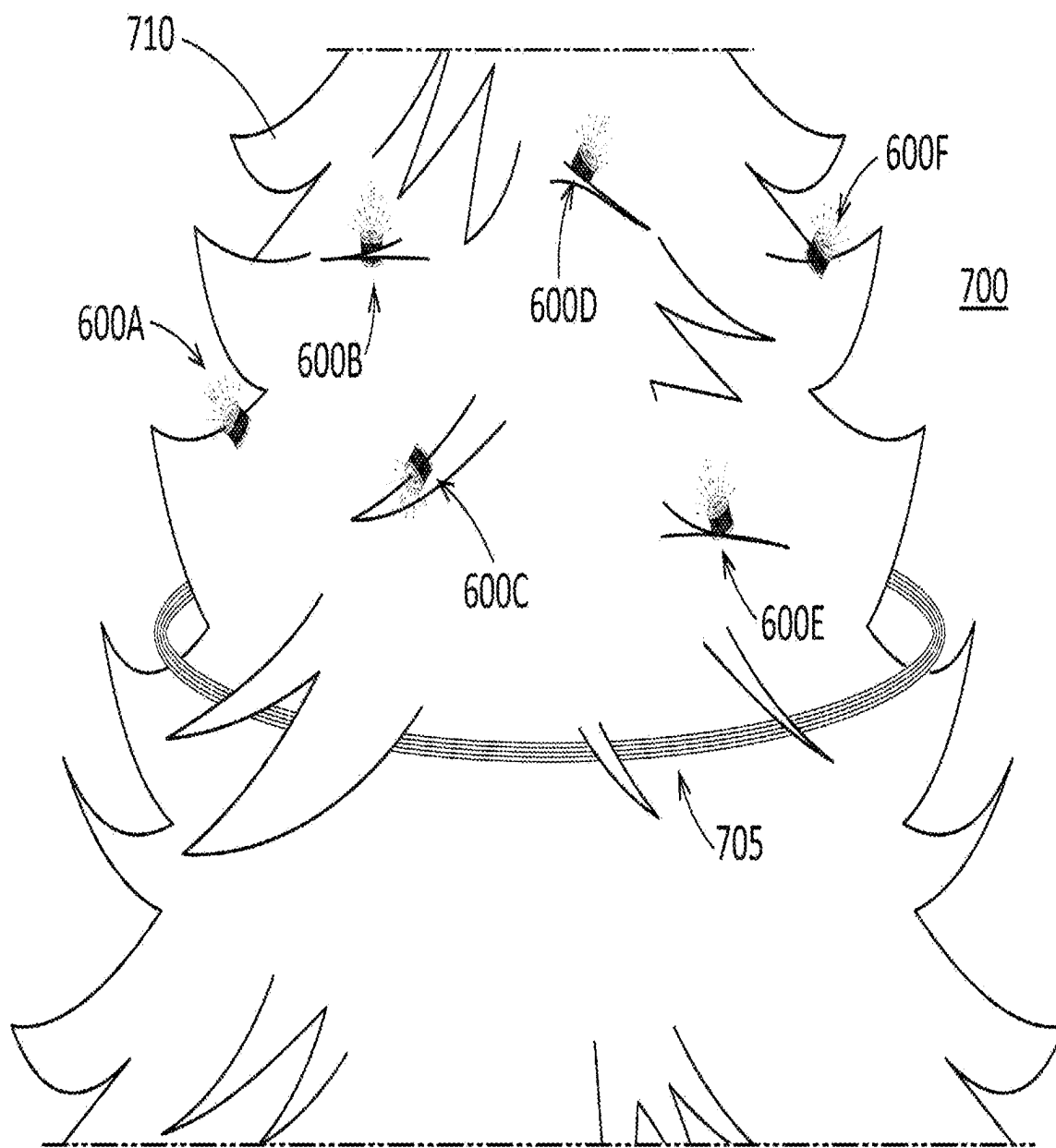


FIG. 7

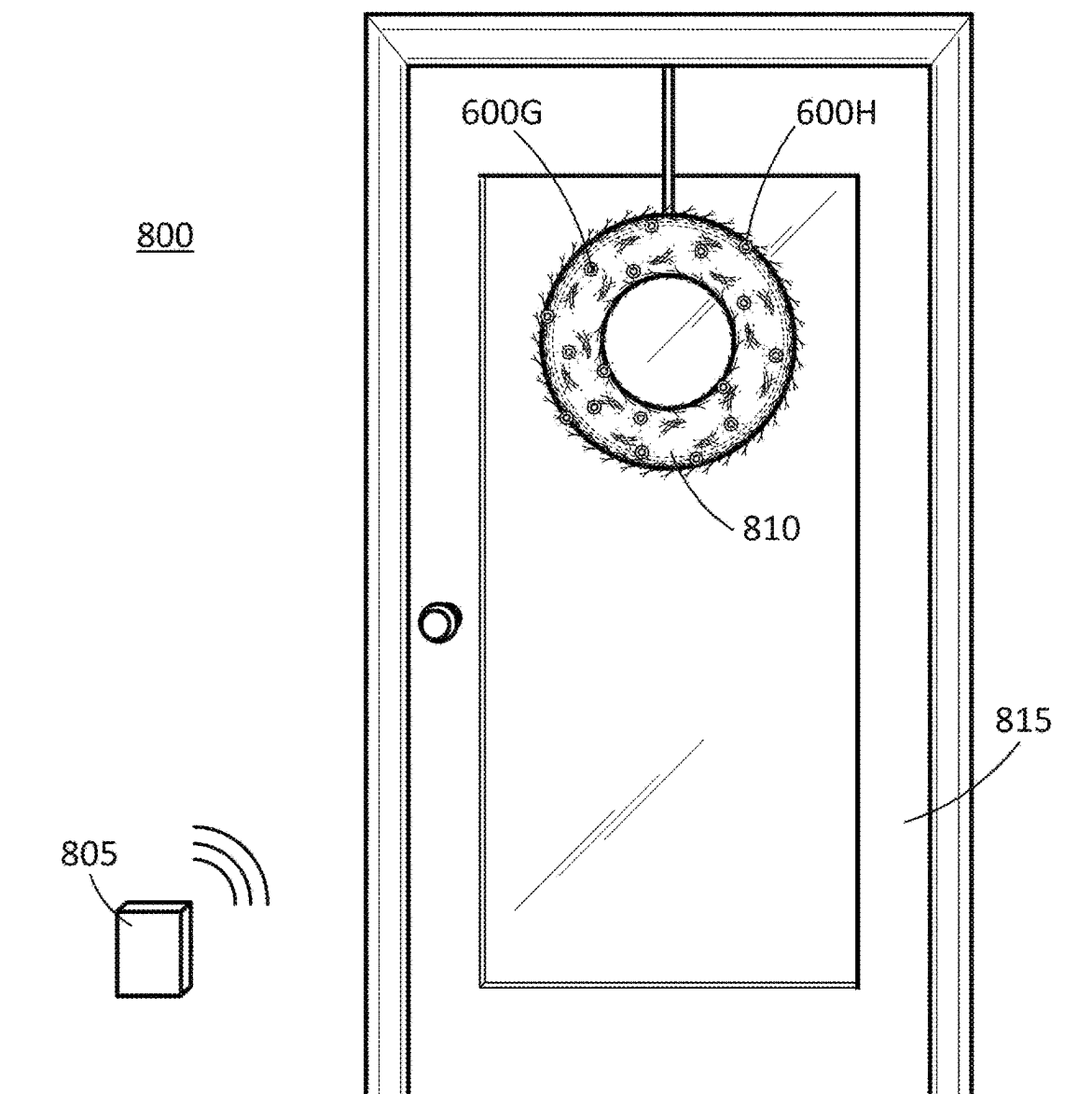


FIG. 8

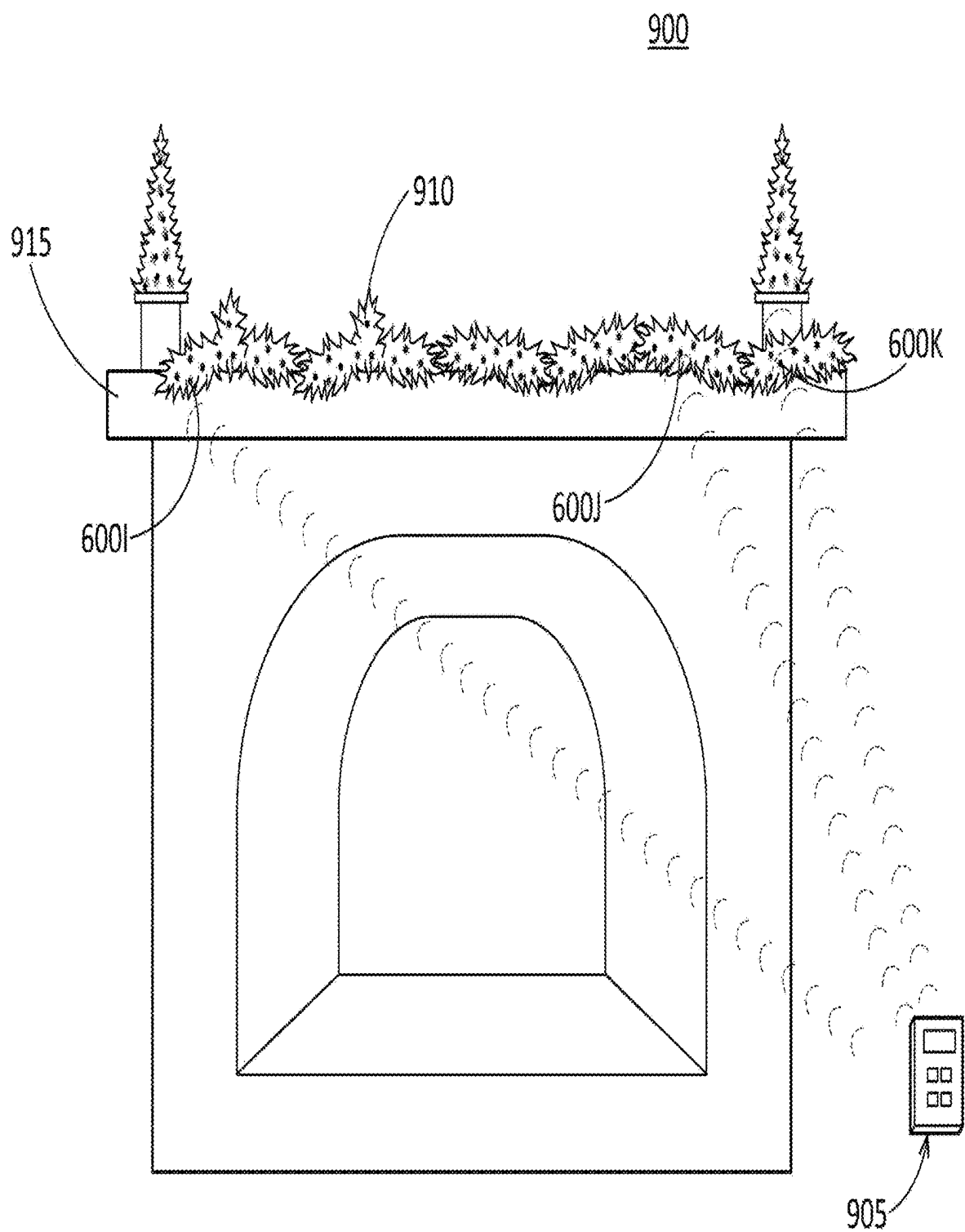


FIG. 9

WIRELESSLY POWERED DECORATIVE SYSTEMS AND DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of U.S. patent application Ser. No. 18/436,686 filed Feb. 8, 2024, which is a continuation-in-part of U.S. patent application Ser. No. 18/325,771 filed May 30, 2023, which is a continuation of U.S. patent application Ser. No. 17/304,103 filed Jun. 14, 2021, each of which are hereby incorporated in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to wirelessly powered decorative objects, such as artificial Christmas trees, wreaths, and ornaments.

BACKGROUND

[0003] Decorative artificial trees are growing in popularity and in turn the types, styles and uses of these trees are multiplying. Usually made to resemble various types of fir or palm trees, or trees without foliage, artificial trees are often decorated and displayed indoors or outdoors to celebrate special occasions, or otherwise displayed to create ambiance.

[0004] Among the trends in artificial trees over the past couple of decades has been the development of controllable lighting systems, which systems require the transmission of energy and data.

SUMMARY

[0005] Embodiments of the present disclosure provide a wirelessly powered decorative system. The wirelessly powered decorative system can include a wreath and a wirelessly powered decorative device in communication with the wreath. The wirelessly powered decorative device can include a receiver and a load. The wirelessly powered decorative system further can include a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device. The load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.

[0006] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0008] FIG. 1 is a diagram illustrating an assembled artificial tree with releasably connected components according to some embodiments of the present disclosure.

[0009] FIG. 2A is a diagram illustrating a trunk portion of the artificial tree according to some embodiments of the present disclosure.

[0010] FIG. 2B is a diagram illustrating the wireless power/data transmission within the artificial tree, according to some embodiments of the present disclosure.

[0011] FIG. 2C is a diagram illustrating another trunk portion of an artificial tree according to some embodiments of the present disclosure.

[0012] FIG. 3A is a diagram illustrating another trunk portion of an artificial tree according to some embodiments of the present disclosure.

[0013] FIG. 3B is a diagram illustrating another trunk portion of an artificial tree according to some embodiments of the present disclosure.

[0014] FIG. 3C is a diagram illustrating another trunk portion of an artificial tree according to some embodiments of the present disclosure.

[0015] FIG. 4 is a diagram illustrating a branch portion of an artificial tree according to some embodiments of the present disclosure.

[0016] FIG. 5 is a diagram illustrating an ornament for the artificial tree, according to some embodiments of the present disclosure.

[0017] FIG. 6 is diagram illustrating a wirelessly powered decorative device 600, according to some embodiments of the present disclosure.

[0018] FIG. 7 is a diagram illustrating a wirelessly powered decorative system 700, according to some embodiments of the present disclosure.

[0019] FIG. 8 is a diagram illustrating a wirelessly powered decorative system 800, according to some embodiments of the present disclosure.

[0020] FIG. 9 is a diagram illustrating a wirelessly powered decorative system 900, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0021] The following disclosure provides many different exemplary embodiments, or examples, for implementing different features of the provided subject matter. Specific simplified examples of components and arrangements are described below to explain the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0022] The terms used in this specification generally have their ordinary meanings in the art and in the specific context where each term is used. The use of examples in this specification, including examples of any terms discussed herein, is illustrative only, and in no way limits the scope and meaning of the disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given in this specification.

[0023] Although the terms “first,” “second,” etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the embodiments. As used herein, the term “and/or” can include any and all combinations of one or more of the associated listed items.

[0024] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0025] In this document, the term “coupled” may also be termed as “electrically coupled”, and the term “connected” may be termed as “electrically connected”. “Coupled” and “connected” may also be used to indicate that two or more elements cooperate or interact with each other.

[0026] Examples of the present disclosure relate to decorative artificial trees, such as Christmas trees, palm trees, Valentine trees, Easter trees, Halloween trees and the like. Although preferred examples of the disclosed technology are explained in detail, it is to be understood that other examples are contemplated. Accordingly, it is not intended that the disclosed technology is limited in its scope to the details of construction and arrangement of components set forth in the following description or illustrated in the drawings. The disclosed technology is capable of other examples and of being practiced or carried out in various ways. Also, in describing the preferred examples, specific terminology will be resorted to for the sake of clarity. It is intended that each term contemplates its broadest meaning as understood by those skilled in the art and can include all technical equivalents which operate in a similar manner to accomplish a similar purpose.

[0027] Wireless transmission of energy can be used to provide power for the lights on the branches and provide for other actions, such as enabling motion of objects on the tree or branches of the tree, or for levitation of objects, such as tree toppers. This can be accomplished by near field or non-radiative wireless transmission, which enables power to be transferred over short distances by magnetic fields using inductive coupling between coils of wire, or by electric fields using capacitive coupling between metal electrodes.

[0028] The present disclosure relates to a system for wirelessly transmitting energy to be used to operate or decorate an artificial tree. Among the benefits of a wireless system for transmitting energy is the elimination of physical copper and plastic wiring. The elimination of copper and plastic wiring reduces costs due to the savings earned by reducing these materials and reducing the cost of labor need to make an artificial tree. The elimination or reduction in wiring also improves the look of the artificial tree by reducing unsightly wires. The elimination of wiring also makes set-up, use, and disassembly easier for the consumer. The elimination of copper and plastic wiring also reduces weight and ultimately, reduces waste. The elimination of

copper and plastic wiring also reduces the potential for operational deficiencies associated with wired connections, including the deterioration of the wires over time or the wires being compromised by mechanical interference.

[0029] The present disclosure identifies providing wireless energy to power lights, or create motion, or otherwise provide power to cause other actions in, on, or near a decorative object like a tree, such as but not limited to levitation of objects enjoyed with or part of the decoration or use of an artificial tree. In one embodiment, the tree uses the inside of a hollow pole or trunk to house a single transmitter or multiple transmitters and a single receiver or multiple receivers and additionally in another embodiment, a single or multiple repeaters, to transmit or transfer and receive wireless energy thru the pole or trunk.

[0030] In another embodiment, the source of the wireless energy may not be a transmitter unit that comes with or is otherwise part of the tree apparatus, but a source external and apart from the tree, such as a separate system able to transmit energy that can be received by various devices or appliances created to receive and use such transmitted power in or throughout a room or other area.

[0031] Reference is made to FIG. 1, which illustrates an artificial tree 100 in communication with a wireless power receiving unit 110 is provided in proximity to the artificial tree 100. The wireless power receiving unit 100 can be a device capable of receiving power wirelessly from wireless power transmitting unit (not shown) that can be provided along with the artificial tree 100 or be separate from the artificial tree 100 and provide power to multiple devices. As illustrated in FIG. 1, the wireless power receiving unit 110 can be configured with a wired connection to the tree; thus, the wireless power receiving unit 110 can receive power via a wireless power transmission and provide primary or supplemental power to the artificial tree 100, including decorative lighting, ornaments, tree toppers, projectors, motion controls, and lighting systems on the artificial tree 100. In other embodiments, the wireless power receiving unit 110 can be configured with a wireless connection to the tree.

[0032] In some embodiments, the power storing unit 120 illustrated in FIG. 1 can be used to store and retain energy obtained by the wireless power receiving unit 110 and to provide energy to the artificial tree 100 for lights, decorations, ornaments, tree toppers, projectors, and other powered components. The power storing unit 120 can be battery, capacitive device, or other device capable of storing charge and or power. In some embodiments, the power storing unit 120 can provide power at different wattage, amperage, or voltage than the wattage, amperage, or voltage at which the power is received by the wireless power receiving unit 110. In some embodiments, the wireless power receiving unit 110 could receive power at a relatively low wattage, provide the energy to the power storing unit 120 for storage, and the power storing unit 120 could provide power at relatively high wattage. Accordingly, the wireless power receiving unit 110 could provide a charge to the power storing unit 120 at relatively low rate, such as a trickle charge, and store that energy for consumption at a higher rate by the artificial tree 100. In a non-limiting example, the wireless power receiving unit 110 could receive power at a relatively low wattage and provide that energy to the power storing unit 120 for storage over a 24-hour period, and the power storing unit 120 could provide a relatively high wattage of power output to the

artificial tree **100** for a much shorter time period, such as one or two hours. In this embodiment, the lighting system of the artificial tree **100** may only be illuminated for a few hours of the day, but the power storing unit **120** is continually storing power received by the wireless power receiving unit **110**. For this embodiment, the electromagnetic radiation or equivalent wireless energy transmission rate in the area adjacent to the artificial tree **100** could be kept advantageously low while enabling the provision of power to the artificial tree **100** at a substantially higher rate. Thus, the configuration of the wireless power receiving unit **110** in communication with the power storing unit **120** is beneficial.

[0033] Reference is made to FIG. 2A, which is a diagram illustrating a trunk portion **200** of the artificial tree **100** according to some embodiments of the present disclosure. As shown in FIG. 2A, the trunk portion **200** can include multiple trunk sections **210**, **220**, and **230**, which can be assembled to collectively form the trunk portion **200**.

[0034] In some embodiments, a power transmission unit can be connected to the trunk portion **200** or base (or separately situated in or around the tree) and can utilize the electricity from an external power source **900** to generate a beam of electromagnetic radiation or equivalent wireless energy. This beam can take the form of visible light, microwave radiation, near infrared radiation or any appropriate frequency or frequencies, depending on the technology chosen for a particular embodiment. The interchangeable component assemblies can include a power reception unit receives power from one or several power transmitters, and converts the total power received to electricity. Similar to the focusing of the transmitted power, it is possible to concentrate the received power for conversion, using receiving arrays, antennas, reflectors or similar means. The power reception unit technology can be a rectenna, a photovoltaic cell, nanotechnology antennas or any other technology capable of converting electromagnetic radiation to electricity, or any combination of the above.

[0035] As shown in the embodiments in FIG. 2A, a transmitter **212** is arranged and located in the lower trunk section **210**. Particularly, in some embodiments, the transmitter **212** can be connected, via an electrical wire, to the external power source **900**. The power source **900** is an available main power supply, which may be a power grid, a solar power system, a battery system, or the like. For example, the transmitter **212** may be connected to the power grid via an electric cord and a wall power socket. In other embodiments, the external power source **900** is a solar power or ambient light or energy harvester system, or other sustainable energy source (e.g., a wind power system), independent of any power grid and used solely to provide power to transmitter **212**.

[0036] The transmitter **212** is configured to convert the electrical power received from the power source **900** into an electromagnetic field by use of the wireless power transmission unit, such as a coil. In some embodiments, the transmitter **212** may include one or more antenna units to transmit power and/or data. For example, the transmitter **212** may have one or more wireless power transmission units **2121**, such as transmitting coils. In addition, the lower trunk section **210** may also include a receiver **214**, which is arranged and located in the lower trunk section **210** and above the transmitter **212**. In some embodiments, multiple receivers **214**, **224**, and **234** can be separately arranged in different trunk sections **210**, **220**, and **230** of the trunk

portion **200**. Receivers **214**, **224**, and **234** are configured to receive the electromagnetic field generated by the transmitter **212**, and then generate corresponding electrical currents accordingly. Thus, the receivers **214**, **224**, and **234** can provide the generated currents to one or more loads connected to the receivers **214**, **224**, and **234**, such as LED lights, ornaments, or tree toppers, such as stars or angels.

[0037] As shown in FIG. 2A, in some embodiments, each trunk section may include one receiver, but the present disclosure is not limited thereto. In addition, in some embodiments, receivers **214**, **224**, and **234** may include one or more openings, passageways, or channels **2143**, which ensures sufficient open passages in the trunk portion such that the electromagnetic field can reach the last or the final receiver at the top trunk section of the trunk portion **200** and allows a greater wireless power transmission efficiency. In various applications, the power consumption required by decorative lights and/or IC components for other lights has been reduced significantly. Therefore, the power level transferred wirelessly, over moderate distances, inside the pole may be within a lower power range, from few watts to several watts.

[0038] In some embodiments, the wireless power transmission units are transmitting coil(s) in the transmitter **212** and receiving coils **2141**, **2241**, **2341** in the receivers **214**, **224**, and **234**, which may be implemented by flat spiral coils formed by wound copper wires. In some embodiments, the transmitter **212** and the receivers **214**, **224**, and **234** may also communicate through inductive coupling via induced magnetic fields between transmitting and receiving coils (loop antennas). Particularly, an alternating current (AC) through the transmitter coil in the transmitter **212** may create an oscillating magnetic field, which first passes through the receiving coil in the receiver **214**, and then the oscillating magnetic field further passes through the receiving coil in the receivers **224** and **234** respectively. Accordingly, alternating voltages are induced in the receiving coils, which generate alternating currents in the receivers **214**, **224**, and **234**. In some embodiments, the induced alternating currents may drive corresponding loads electrically connected to trunk sections **210**, **220**, and **230** directly. In some other embodiments, the induced alternating currents can be rectified by a rectifier in the receivers **214**, **224**, and **234** to direct current (DC) to drive the loads.

[0039] For example, the wireless power transmission can be used to provide power for the LED lights on the branches of the artificial tree, without having to use physical couplers between the wiring in or along the tree trunk (or pole) and the branches of the artificial tree. As shown in FIG. 1, each of the receivers **214**, **224**, and **234** may include one or multiple output connectors. The output connectors are configured to be connected to the loads at the branches and enable transmission of the power to the loads located at the branches. In some embodiments, the trunk sections **210**, **220**, and **230** are drilled to create holes for mounting these output connectors. Accordingly, in some embodiments, the branches provide wired LED lights but the trunk portions are connected wirelessly. In other embodiments, all connections are wireless.

[0040] As an example, a trunk section can include a plurality of releasably connected pre-lit branch assemblies configured to be mechanically attachable and detachable to a trunk body of the trunk section and electrically attachable

and detachable to the main power source of the tree (e.g., power source, non-radiating wireless energy field, etc.).

[0041] As a further example, a trunk section can include one or more releasably connected wiring harnesses configured to electrically connect the lighting elements of a releasably connected pre-lit branch assembly to the main power within or along a trunk section. As will be appreciated, such wiring harnesses allow for ease of connection and disconnection while also providing flexibility for the overall tree assembly. For example, wiring harnesses can be designed to allow for various types of power (e.g., low voltage systems for powering LEDs or small electronics, medium voltage systems for providing power for a large array of LEDs or small electronics, high voltage systems originating from a wall outlet, etc.) and lighting configuration (e.g., series wired lighting elements, parallel wired lighting elements, series wired light strings, parallel wired light strings, parallel series wired light strings, series parallel wired light strings, etc.). Such design flexibility benefits both the consumers as they have the ability to easily maintain and customize their tree, but also to the manufacturers as their ability to manufacture a batch of releasably connected components can lead to vast array of easily assembled artificial trees. Additionally, electrical connection can carry power, data, or both. For example, an electrical connection can facilitate the powering lights on a pre-lit branch assembly. An electrical connection can facilitate the transmission of a control signal operable to, for example, change lighting settings of pre-lit branch assemblies.

[0042] In some embodiments, the mechanical connection of the tree poles can be achieved by a plastic mechanical connection. As shown in FIG. 2A, each of the trunk sections **210**, **220**, and **230** can be configured to set into another trunk section via a base part with a sprocket structure or a clutch structure. The sprocket structure or the clutch structure is provided for preventing the rotation after two corresponding trunk sections are assembled.

[0043] Reference is made to FIG. 2B, which is a diagram illustrating the wireless power/data transmission within the artificial tree **100**, according to some embodiments of the present disclosure. As discussed above, the transmitter **212** connected to the pole or base of the artificial tree **100**, or separately situated in or around the tree, can utilize electromagnetic radiation. Use of wireless transmission reduces the connections and wirings needed to transmit electrical signals (such as power or data) throughout the artificial tree **100**. In some embodiments, the transmission can be visible light, microwave radiation, near infrared radiation or any appropriate frequency or frequencies, depending on the technology chosen. For example, wireless transmission of energy can be used to provide power for the lights on the branches without having to use connectors between wiring in or along the pole or tree trunk and the branches.

[0044] As depicted in FIG. 2B, the lower trunk section **210** can include the wireless transmitter **212**. The wireless transmitter **212** is operable to transmit power or data signals to a plurality of receivers **214**, **224**, and **234**. In some embodiments, the wireless transmitter **212** does not transmit power and only transmits data; thus, the plurality of receivers **214**, **224**, and **234** are only used to receive data. As discussed above, in other embodiments the wireless transmitter **212** transmits both power and data. In those embodiments, some receivers only receive data, others only receive power, and others receive power and data. Those of skill in the art will

appreciate that the data transmitted can be used to communicate with or control various devices on the artificial tree **100**, such as decorative lighting, ornaments, tree toppers, projectors, motion controls, and lighting systems. The receivers **214**, **224**, and **234** can be part of the trunk section **210**, **220**, and **230**, or part of the plurality of releasably connected pre-lit branch assemblies, power distribution hubs, other portions of the artificial tree **100**, or any combination thereof. Similar to the focusing of the transmitted power, it is possible for the plurality of receivers to concentrate the received power for conversion, using receiving arrays, antennas, reflectors or similar means. Additionally, or alternatively, a power distribution hub may include a wireless transmitter, receiver, transceiver, or a combination thereof. As will be appreciated, any one or more of the trunk sections **210**, **220**, and **230** and the branch portion **400**, may include transmitters, receivers, or transceivers, to transmit or receive signals controlling or powering the brightness or effect of the components (e.g., LED lights) installed in the branch portion **400** or the ornament(s) **500**. In some embodiments, a star tree topper **510** of the artificial tree **100** in FIG. 2B may also be a wireless ornament capable of being powered wirelessly.

[0045] Reference is made to FIG. 2C, which is a diagram illustrating another trunk portion **200** of an artificial tree according to some other embodiments of the present disclosure. As shown in FIG. 2C, in some embodiments, the trunk section **210** can include one transmitter **212**, and the trunk section **220** can include one receiver **224**. Accordingly, the transmitter **212** can receive power from the external power source **900** and transmit the power wirelessly to the receiver **224** in the next trunk section **220**. Structures and operations of the transmitter **212** and the receiver **224** have been discussed in detail above and thus are not repeated herein for the sake of brevity.

[0046] Reference is made to FIG. 3A, which is a diagram illustrating another trunk portion **200** of an artificial tree according to some embodiments of the present disclosure. As shown in FIG. 3A, in some embodiments, each of the trunk sections **210**, **220**, and **230** may include at least one transmitter (e.g., transmitters **212**, **222**, and **232**), and at least one receiver (e.g., receivers **214**, **224**, and **234**). In addition, in some embodiments, each of the trunk sections **210**, **220**, and **230** may further include battery components **216**, **226** and **236** respectively coupled to the receivers **214**, **224**, and **234** and configured to be charged using the power outputted from the receivers **214**, **224**, and **234**. Accordingly, transmitters **222**, and **232** can respectively receive power from the battery components **226** and **236**, and further transmit the power wirelessly to the next trunk section. Furthermore, in some embodiments, each of the trunk sections **210**, **220**, and **230** may include one or more intermediate resonators **228** (repeaters) placed between corresponding transmitter(s) and receiver(s). The repeaters **228** arranged within the trunk sections **210**, **220**, and **230** are configured to enhance the transfer distance of the wireless power.

[0047] As shown in FIG. 2A and FIG. 3A, in some embodiments, transmitting and receiving coils may be disposed at a center (or core) part of the trunk sections, with openings **2143** between the centrally located coil and an outside rim of the receiver, but the present disclosure is not limited thereto. In some embodiments, transmitting and receiving coils may be disposed contiguous to the outside rim of the receiver, and an open passageway is provided at

the center (or core) part of the trunk sections. By locating transmitting and receiving coils in each trunk sections **210**, **220**, and **230** on a common axis and providing unobstructed pathways or channels, the magnetic field generated by one coil is able to pass through another coil properly, which enhances the wireless transmission efficiency.

[0048] Reference is made to FIG. 3B, which is a diagram illustrating another trunk portion **200** of an artificial tree according to some embodiments of the present disclosure. As shown in FIG. 3B, in some embodiments, the artificial tree **100** is a single pole artificial tree with one trunk section **210**. Those of skill in the art will appreciate that the trunk section **210** can be made up of multiple physical pieces that can be assembled by the user. The trunk section **210** can include a single transmitter **212** and a single receiver **214**. Accordingly, the transmitter **212** can receive power from the external power source **900**, and transmit the power wirelessly to the receiver **214** in the same trunk section **210**. Structures and operations of the transmitter **212** and the receiver **214** have been discussed in detail above and thus are not repeated herein for the sake of brevity.

[0049] Reference is made to FIG. 3C, which is a diagram illustrating another trunk portion **200** of an artificial tree according to some embodiments of the present disclosure. As shown in FIG. 3C, in some embodiments, the artificial tree **100** is a single pole artificial tree with one trunk section **210**. The trunk section **210** only can include a single receiver **214**. The receiver **214** is configured to generate an output current in response to an electromagnetic field transmitted to the trunk section **210** from an external power source. In some embodiments, the artificial tree **100** may be powered wirelessly. For example, the receiver **214** can receive wireless power from a generic transmitter **310**, which is external from the trunk section **210**. In some embodiments, the generic transmitter **310** may be arranged within another part of the artificial tree **100**. In some other embodiments, the generic transmitter **310** may be arranged apart from the artificial tree **100** and connected to the external power source **900**, and configured to transmit the power wirelessly to the receiver **214** in the trunk section **210**. In some other embodiments, lighting elements within the artificial tree **100** may be powered by the output current via internal wired connections coupling the receiver **214** and the lighting elements. Structures and operations of the transmitter **310** and the receiver **214** have been discussed in detail above and thus are not repeated herein for the sake of brevity.

[0050] In various embodiments, the distances of the coils, the necessary and allowable signal frequency inside the pole may be designed based on actual needs. In some embodiments, the coil size, including at least the flat single layer coils illustrated in FIG. 2A and FIG. 3A may be determined based on the diameter of the trunk sections **210**, **220**, and **230**. In addition, the pole formed by the trunk sections **210**, **220**, and **230** may be a hollow metal, aluminum, or polymer pole, which enables energy to be contained and directed within the confines of the metal, aluminum, or polymer tube. The metal, aluminum, or polymer pole may achieve better safety and/or efficiency performance, but the present disclosure is not limited thereto.

[0051] Accordingly, by implementing wireless power transmission within a single trunk **210** or between different trunk sections **210**, **220**, and **230**, the physical conductors and/or connectors needed for electrical or data transmission within or around a pole can be eliminated. Therefore, a

simplified and smooth mechanical connection for connecting the trunk sections **210**, **220**, and **230** can be realized for assembling an artificial tree with power supply nodes at both lower portions and upper portions of the tree. Therefore, the installation of the artificial tree can be simplified, and the weight of the artificial tree can be reduced because at least some portions of the wiring can be removed. In addition, the unit costs associated with artificial trees are also reduced.

[0052] Reference is made to FIG. 4, which is a diagram illustrating a branch portion **400** of an artificial tree according to some embodiments of the present disclosure. In some embodiments, the branch portion **400** can be a releasably connected pre-lit branch assembly electrically connected to the artificial tree **100**. As depicted, the branch portion **400** can include a main branch **405** and a plurality of attached sub-branches **410**. As further depicted, the main branch **405** can include a trunk connector assembly **425**. Trunk connector assembly **425** can be configured to attach and detach to the artificial tree **100** and ensure a stable connection when attached. The trunk connector assembly **425**, as depicted, is in an unassembled state.

[0053] In some embodiments, the branch portion **400** may be electrically connected, via the output connectors located on the receivers, to one of the trunk sections **210**, **220**, and **230**, but the present disclosure is not limited thereto. For example, similar to the wireless power transmission between trunk sections **210**, **220**, and **230**, the branch portion **400** may include one or more receivers for wireless power transmission from trunk sections **210**, **220**, and/or **230** to the branch portion **400**.

[0054] The pre-lit branch assembly as well as other releasably connected components discussed here may be releasably connected in several ways. In some embodiments, the mechanical connection is separate and independent from the electrical connection and the mechanical connection and the electrical connection can be connected separately. For example, the mechanical connection can include a removable fastener assembly for connecting and detaching the branch assemblies from the tree pole including a clevis, a clevis pin, and a cotter pin, while the electrical connection can include a plug and receptacle pair or wirelessly connection (or both). In some embodiments, the electrical and mechanical connections are dependent upon each other.

[0055] For example, a connector may include both an electrical connector and a mechanical connector (e.g., mechanical pins and receptacles and electrical plugs and receptacles). In other embodiments, the connection may be an electromechanical connection that provides both a mechanical and electrical connection. For example, plug and receptacle pairs may establish electrical and mechanical connection such that the pre lit branch assembly is releasably attached to another component such as a trunk and receives electrical signal from the component. In some embodiments, the electrical connection occurs wirelessly. The wireless electrical connection may be used to transmit data, power, or both.

[0056] There may be multiple electrical and/or mechanical connections. For example, a pre-lit branch assembly may receive power from a wireless connection and data from a wired connection or receive power from a wired connection and data from a wireless connection. The wireless electrical connection may occur as a result of the mechanical connec-

tion (e.g., the mechanical connection brings the branch assembly in range of a wireless signal to thereby establish a wireless connection).

[0057] In some embodiments, the electrical connection may be established indirectly. For example, the branch assemblies are electrically connected to a harness which is electrically connected to the trunk thereby establishing an indirect electrical connection between the branch assemblies and the trunk).

[0058] Also, as shown, the branch portion **400** can include a light string **415** and an electrical connector **420** associated with the one of the output connectors of the tree trunk of the artificial tree **100** while ensuring a stable connection when attached. The electrical connectors may be a plug and receptacle pair, sometimes referred to as a lamp lock assembly. As will be appreciated, many connector types can be used for electrically connecting a releasably connected pre-lit branch assembly to a trunk section (e.g., trunk section **210**, **220**, or **230**) of the artificial tree **100**. The electrical connectors described herein may carry power, data, or both.

[0059] In some embodiments, the branch portion **400** can include a single light string **415** disposed on the branches **405**, **410**. The single light string **415** can include an electrical connector **420** configured to mate with a connector built into the trunk assembly (e.g., an output connector located on the receiver). In some other examples, the branch portion **400** can include multiple branches having one or more light strings **415** disposed across the branches and at least one of the light strings **415** can include an electrical connector configured to mate with an electrical connector built into the trunk assembly.

[0060] In some other examples, the branch portion **400** having the light string **415** with the electrical connector **420** is configured to wirelessly receive power from the trunk assembly power source and distribute the power throughout the light string. Further, instead of the light string **415**, in some cases, the branch portion **400** can include a plurality of individual light emitters that are each configured to wirelessly receive power from the wireless power transmitted from the trunk portion **200** of artificial tree **100**.

[0061] Particularly, the branch portion **400** may also include one or more receivers. Accordingly, the energy required for driving components (e.g., LED lights) installed in the branch portion **400** can be transmit up and across the tree area via the wireless power transmission in the trunk portion **200** to the receiver(s) in the branch portion **400**. Similarly, in some embodiments, the branch portion **400** may also include one or more transmitters configured to transmit the wireless power and/or signals to other devices. For example, in some embodiments, the transmitter(s) in the branch portion **400** may be configured to supply power required by one or more ornaments hung on the branch portion **400**. As will be appreciated, such embodiments further reduce the need for at least portions of the wirings for the pre-lit branch assembly and the unit costs associated with artificial trees can be further reduced. As will be further appreciated, the incorporation of the wireless power decreases the need for physical connecting and disconnecting of lighting during both manufacturing and maintenance of artificial trees.

[0062] Reference is made to FIG. 5, which is a diagram illustrating an ornament **500** for the artificial tree **100**, according to some embodiments of the present disclosure. As shown in FIG. 5, the ornament **500** may be a snow globe

to be hung on the artificial tree **100** (or displayed in various ways). In some embodiments, a control unit, a lighting module and/or a motion module, can be integrated in the ornament **500**. For example, the control unit can be electrically connected to one or more LED components and a micro electric motor, and configured to output driving signals to drive the LED component(s) and the micro electric motor, in order to create an illuminated and/or blustering snow scene within the ornament **500**. The ornament **500**, including the circuits and electrical components within the ornament **500**, can be powered wirelessly and displayed when hung on the artificial tree.

[0063] It would be appreciated that while in FIG. 5, the ornament **500** is in the form of a snow globe, the present disclosure is not limited thereto. For example, in some embodiments, the ornament **500** can be a LED light source, or other decorative items (e.g., candy canes) configured to be illuminated internally with LED lighting component(s) arranged in the ornament **500** and powered wirelessly by the receiver(s) integrated in the ornament **500**. For example, various lighting patterns can be applied the decorative candy canes, such as alternating red and white stripes, or motion can be caused, but the present disclosure is not limited thereto. The red and white stripes of the decorative candy canes may be created by diagonal rows of alternating red and white LED units respectively. Accordingly, the power can be wirelessly transferred from the trunk portion **200** and/or the branch portion **400** to the LED light sources, the decorative candy canes, or other decorative items hung on the artificial tree by corresponding transmitter(s) and receiver(s) to supply the electricity for illuminating the ornament **500**, projecting light or images, or causing motion to occur as part of the ornamental display.

[0064] In some other embodiments, the ornament **500** may be a tree topper, such as a star or angel. Similarly, the tree topper can be illuminated, wirelessly, with various programmed lighting effects, such as a flickering or sparkling light effect, or a color changing lighting effect, or may levitate. For example, a star tree topper may be configured to glow or pulse like a real star by programed LEDs, alternating between different brightness level. For example, the star tree topper may be switched gradually between a dimmer light and a brighter light repeatedly.

[0065] In some other embodiments, the ornament **500** is in the form of one or more decorative candles made to look like old-fashioned wax candle(s) that were the original lighting for the Christmas trees. Similarly, the ornament **500** can include one or more LED units configured to mimic the flickering of a real candle flame and is powered wirelessly by the receiver(s) integrated within the ornament **500**. For example, the LED flickering flame can be created by a series of programed LEDs, providing a flickering light effect. It would be appreciated that, in some other embodiments, the ornament **500** may be other ornaments in forms or shapes commonly used as ornaments for display on decorative trees for one skilled in the art, or may be an imbedded light source in the foliage or branches of the artificial tree **100**.

[0066] As described above, the receivers or transceivers can be arranged and integrated in lighting units, such as LED sources, various objects or ornaments, or embedded within the stem of a branch portion of the artificial tree **100**. In some embodiments, the receivers or transceivers in the artificial tree **100** may also be configured to receive wireless signals from other transmitters that are not part of the artificial tree

100. For example, the receivers or transceivers integrated in the ornaments may also receive signals from a power supplying device located in or around a dwelling for transmitting energy and/or data wirelessly within a given area or region. In some alternative embodiments, the receiver (e.g., receivers **214**, **224**, or **234**) installed in one or more trunk sections **210**, **220** or **230** of the trunk portion **200** may also be configured to receive an electromagnetic field transmitted to the trunk section from an external wireless power source that is not part of the artificial tree **100**, and configured to generate the output current in response to the received electromagnetic field signals. Accordingly, the lighting module coupled to the trunk can receive the output current from the receiver, so that one or more LED components in the lighting module can be powered by the output current.

[0067] In some other embodiments, the artificial tree **100** may also be used as a base or hub station for interacting with other units or devices and providing energy and/or data wirelessly to the units or devices around the artificial tree **100**. For example, the artificial tree **100** may be physically connected to the power source via wires and configured to provide energy and/or interact with other decoration items, such as wreaths/garlands, mini-trees/secondary trees, tree toppers, a Christmas train set, a nativity scene (also known as a manger scene, crib, crèche), a monitor or a display, a speaker, an aroma device, a smoke emitting device (for Halloween trees), or the like. In some embodiments, additional battery device(s) can be arranged external to the pole of the artificial tree **100** and located within a hub system located outside of the pole but near to the tree. The battery device(s) can be configured to receive and store energy from wireless source(s) located inside the house (or other building), and then to supply the power required by the artificial tree **100**.

[0068] Reference is made to FIG. 6, which is a diagram illustrating a wirelessly powered decorative device **600**, according to some embodiments of the present disclosure. The wirelessly powered decorative device **600** is enabled to receive a wireless power transmission. In the embodiment depicted in FIG. 6, the wirelessly powered decorative device **600** can include a receiver **605**. Receiver **605** can be configured to receive an electromagnetic field generated by a transmitter and then generate corresponding electrical currents accordingly. In some embodiments, receiver **605** can provide the generated currents to one or more loads connected to the receivers **605**. In some embodiments, the wirelessly powered decorative device **600** is provided in a wirelessly powered decorative system **700** that can include a wireless power transmitter, such as a transmitter that can be powered by a wired power source or battery power source. In other embodiments of the wirelessly powered decorative system **700**, a transmitter is not included in the system and the wirelessly powered decorative device **600** is configured to receive wireless power transfer from an external source.

[0069] In the embodiment depicted in FIG. 6, the wirelessly powered decorative device **600** can include a load **610** such as LED device. In some embodiments, receiver **605** can receive an electromagnetic field generated by a transmitter and then generate a corresponding electrical current to power the load **610**, such as illuminating an LED device. Those of skill in the art will appreciate that the wirelessly powered decorative device **600** can be a single unit, such that receiver **605** and load **610** are contained within the same unit. In alternative embodiments, the wirelessly powered

decorative device **600** can be multiple independent units that are in communication with each other, either wired or wireless communication. For example, the receiver **605** of the wirelessly powered decorative device **600** could be provided in a separate housing from the load **610** and the two components could be connected by a wire.

[0070] As shown in the embodiment in FIG. 6, receiver **605** can be configured to communicate with a wireless transmitter through inductive coupling via induced magnetic fields between transmitting and receiving coils (loop antennas). In some embodiments, an alternating current (AC) through the transmitter coil in the wireless transmitter may create an oscillating magnetic field, which is transmitted to and passes through the receiving coil in the receiver **605**. Accordingly, alternating voltages are induced in the receiving coil, which generates alternating currents in the receiver **605**. In some embodiments, the induced alternating current may drive corresponding loads electrically connected to receiver **605**. In the in the embodiment depicted in FIG. 6, load **610** is provided which incorporates an LED device. In some other embodiments, the induced alternating currents can be rectified by a rectifier in the receiver **605** to direct current (DC) to drive the loads. For example, in the embodiment depicted in FIG. 6, the wirelessly powered decorative device **600** can be configured with a rectifier to convert the alternating current received by the receiver **605** to a direct current and that direct current can be provided to load **610** or a battery. In one embodiment, a DC power source is provided to load **610** via a wireless power transmission, such as a Radio Frequency (RF) transmission. In some embodiments, the receiver **605** of the wirelessly powered decorative device **600** can be configured to receive power transfer via a radio frequency power transfer, capacitive power transfer, or laser power transfer. Furthermore, in some embodiments, the transmitter of the wirelessly powered decorative device **600** can be provided with a timer such that it transmits power only during desired times.

[0071] Those of skill in the art will appreciate that the wirelessly powered decorative device **600** can be deployed in a variety of embodiments. For example, a group of wirelessly powered decorative devices **600** could be removably configured on a decorative object and provided as the lighting elements on a decorative object, such as a wreath, garland strand, artificial tree, ornament, snow globe, decorative inflatable object, mini-artificial tree, tree topper, and home decorative accent. The wirelessly powered decorative devices **600** could be incorporated into decorative objects in Christmas decorations, Easter Decorations, Halloween decorations, New Years decorations, July 4th decorations, and/or decorations for all types of holidays, seasons, events and decors. Furthermore, some embodiments of the decorative objects including the wirelessly powered decorative devices **600** could be used for decorations for multiple holidays or seasons.

[0072] Some embodiments of the wirelessly powered decorative devices **600** can include a battery for the storage of power received from the wireless power transmitter. Those of skill in the art will appreciate that the ability of the wirelessly powered decorative devices **600** to store power, even if only temporarily, enables usage of power for display asynchronously to receiving wireless power transmissions. For example, a wirelessly powered decorative device **600** could receive intermittent wireless transmissions of power or receive continuous wireless power transmissions of rela-

tively low amounts of power and store the received energy in a battery included with the wirelessly powered decorative device 600 or in connection with the wirelessly powered decorative device 600 to be used at a later time. Accordingly, the rate at which power is received for such embodiments of the wirelessly powered decorative device 600 could be significantly less than the rate at which power is consumed by the wirelessly powered decorative devices 600. In some embodiments, the wireless power transmissions could be limited to certain times of the day or when certain conditions are met.

[0073] In many implementations of the present disclosure, the wirelessly powered decorative devices 600 is only consuming power for a small period of the day for display but can receive and store power throughout the day. In some embodiments of the present disclosure, the wirelessly powered decorative device 600 is provided as part of a wirelessly powered decorative system, which can include a wireless power transmitter for which the transmission rates and time periods for transmission are programmable. In one example, wirelessly powered decorative system enables the user to program the wireless power transmitter to only transmit power to the wirelessly powered decorative device 600 from 1:00 AM to 5:00 AM. Accordingly, embodiments of the wirelessly powered decorative system can enable the user to mitigate concerns regarding Electric and Magnetic Fields (EMFs) associated with wireless power transmission.

[0074] In an alternative embodiment of the present disclosure, the wireless power transmitter can be configured to operate in coordination with data received from one or more sensors. In this alternative embodiment, the sensors, such as an optical sensor, audio sensor, or thermal sensor, could be used to determine when the area around the wirelessly powered decorative system is occupied by a human or pet. Thus, in this alternative embodiment, the wireless power transmitter could transmit power only when the sensors have verified that the area around the wirelessly powered decorative system is not currently occupied by a human or pet. This would provide an alternative method for mitigating concerns regarding Electric and Magnetic Fields (EMFs) associated with wireless power transmission. In an additional embodiment of the present disclosure, the sensors of the wirelessly powered decorative system can enable energy efficient operation of the wirelessly powered decorative system. For example, the wirelessly powered decorative system could be configured to only illuminate the wirelessly powered decorative devices 600 when a user is proximate to the wirelessly powered decorative system or when a particular user is proximate to the wirelessly powered decorative system. Additionally, in some embodiments, the display sequence of the wirelessly powered decorative devices 600 can be associated with particular users such that when a particular user enters the vicinity of the wirelessly powered decorative system, the wirelessly powered decorative devices 600 activate a particular display sequence. Alternatively, the wireless power transmitter can be configured with a timing switch such that the schedule for wireless power transmission by the wireless power transmitter 705 can be preprogrammed by the user. Those of skill in the art will appreciate that the wireless power transmitter 705 can be configured to receive mains power via a fault protection circuit, such as a Ground-Fault Circuit Interrupter (“GFCI”) circuit.

[0075] Reference is made to FIG. 7, which is a diagram illustrating a wirelessly powered decorative system 700, according to some embodiments of the present disclosure. As shown in the embodiment in FIG. 7, the wirelessly powered decorative system 700 can be implemented on artificial tree 710. Furthermore, as shown in FIG. 7, the wirelessly powered decorative system 700 can include numerous wirelessly powered decorative devices, such as wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F. As shown in the embodiment in FIG. 7, the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F each include loads having LED devices which can provide wirelessly powered lighting elements for artificial tree 710. As shown in the embodiment in FIG. 7, the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F can be independently placed on artificial tree 710. Thus, in accordance with an embodiment of the present disclosure, the wirelessly powered decorative system 700 can be customized and configured by the user, such that the user may place each of the wirelessly powered decorative devices in a desired location. Those of skill in the art will appreciate that power consumption of LED devices, such as the loads having LED devices in the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F, is relatively low in comparison to other types of lighting devices; thus, the amount of energy that must be wirelessly transmitted is relatively low. For example, the power consumption of 50 LED devices in an exemplary configuration, is in and around 1 watt (1 W), as one 3 volt LED device consumes 0.015 watts when illuminated.

[0076] As shown in the embodiment in FIG. 7, the wirelessly powered decorative system 700 can include a wireless power transmitter 705. Those of skill in the art will appreciate that wireless power transmitter 705 can be configured for near field or non-radiative wireless transmission, which enables power to be transferred over short distances by magnetic fields using inductive coupling between coils of wire, or by electric fields using capacitive coupling between metal electrodes. Alternatively, wireless power transmitter 705 can be configured for radio frequency power transfer, such that energy is broadcast in a radio frequency signal. Those of skill in the art will appreciate that wireless transmission of energy, as disclosed herein, can be achieved using an inductive power transfer (“IPT”), capacitive power transfer (“CPT”), laser power transfer (“LPT”), or radio frequency power transfer (“RFPT”). In the embodiment shown in FIG. 7, the wireless power transmitter 705 can be configured at the base of the artificial tree 710. Those of skill in the art will appreciate that the wireless power transmitter 705 can be configured for transmission over a certain distance. In one embodiment of the present disclosure, the wireless power transmitter 705 can transmit a 1-watt electrical signal the distance of 2 to 3 meters. In an alternative embodiment of the present disclosure, wireless power transmitter 705 can be configured transmit a 0.5-watt electrical signal the distance of 5 to 6 meters. In another embodiment of the present disclosure, wireless power transmitter 705 can be configured transmit a 2-watt electrical signal the distance of less than 1 meter. Those of skill in the art will appreciate that the configuration of the wirelessly powered decorative system 700 can be customized based upon the specific characteristics of the wireless power transmitter 705, including a transmitter for inductive power transfer (“IPT”),

capacitive power transfer (“CPT”), laser power transfer (“LPT”), or radio frequency power transfer (“RFPT”).

[0077] The wireless power transmitter 705 can transmit power to each of the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F on the artificial tree 710. In the embodiment shown in FIG. 7, the wireless power transmitter 705 can be comprised of multiple rings, transmitting coils, that encircle artificial tree 710 proximate to its base. In other embodiments, the wireless power transmitter 705 can be comprised of multiple transmitting coils that are separate and independent and placed in different locations on the artificial tree 710. In some embodiments, the wireless power transmitter 705 is configured to convert the electrical power received from a power source into an electromagnetic field by use of a wireless power transmission unit, such as a coil. For example, the wireless power transmitter 705 may have one or more wireless power transmission units, such as transmitting coils. In other embodiments, the wireless power transmitter 705 may include one or more antenna units to transmit power and/or data. In addition, wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F on the artificial tree 710 can include receivers to receive the transmitted power and/or data. In some embodiments, the receivers are configured to receive the signal generated by the wireless power transmitter 705, and then generate corresponding electrical currents accordingly for consumption by a load, such as an LED device, or to be stored by a battery. Thus, the receivers can provide the generated currents to one or more loads connected to the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F, such as LED lights, ornaments, or tree toppers, such as stars or angels. In some embodiments, the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F can be configured to send a beacon signal to locate a wireless power transmitter, such as wireless power transmitter 705. In some embodiments, the wireless power transmitter 705 is configured to transmit power to the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F in designated transmission paths and the communication between wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F and wireless power transmitter 705 can be nearly instantaneous. In some embodiments of the present disclosure, each of the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F can include a receiver for receiving wireless power transmissions from the wireless power transmitter 705. In other embodiments of the present disclosure, the receiver can be separate from the wirelessly powered decorative devices and provide power to one or more of the wirelessly powered decorative devices. Accordingly, in some embodiments, the artificial tree 710 has one or more receivers placed in convenient positions for receiving power transmission and the wirelessly powered decorative devices 600A, 600B, 600C, 600D, 600E, and 600F are placed in desired positions for decorative purposes.

[0078] Reference is made to FIG. 8, which is a diagram illustrating a wirelessly powered decorative system 800 according to some embodiments of the present disclosure. As shown in the embodiment in FIG. 8, the wirelessly powered decorative system 800 can be implemented on decorative wreath 810. Those of skill in the art will appreciate that decorative wreath 810 is one embodiment of a decorative object that can be deployed along with the

wirelessly powered decorative system 800 and that the decorative object could include garland stands, holiday swag, decorative foliage, decorative flowers, decorative berries, tree-toppers decorative Christmas baubles, decorative Easter eggs, and decorative objects for many other types of holidays, parties, and events. The embodiment of the wirelessly powered decorative system 800 shown in FIG. 8 enables the decorative wreath 810 to be placed on door 815 without the need for a wired connection to a power source or large battery-based power source on the wreath 810. As those of skill in the art will appreciate, doors, like door 815, are not typically configured with an integrated power source; thus, wiring a decorative system on decorative wreath 810 mounted on door 815 can be problematic because it may interfere with the operation of door 815 and/or require wiring that is not aesthetically pleasing and may detract from the desired decorative look. Thus, if a user desires to add lights to a decorative object it can be problematic because the electric cords will be visible or there is not a convenient socket. Additionally, conventional battery-based power sources for decorative objects, like decorative wreath 810, are known by those of skill in the art to have numerous drawbacks as such conventional battery-based power sources for decorative systems can be large and cumbersome, detract from the decorative aesthetic, and the batteries must be replaced frequently and often corrode. Further negatives to using batteries for decorative objects are known to those of skill in the art, including that the battery life may not be sufficient, requiring extra work changing batteries during the desired term of usage, and batteries often result in waste, waste that can be harmful due to the materials inside batteries. One advantage of the decorative system 800 of the present disclosure is that it does not require large batteries or cumbersome power cords but instead provides energy wirelessly to the decorative system 800 incorporated with wreaths, garlands, swags, artificial trees, or other decorative objects, thereby enabling display of these decorative objects in all customary locations.

[0079] In accordance with embodiments of the present disclosure, wirelessly powered decorative system 800 can be configured to wirelessly provide power to wirelessly powered decorative devices deployed on decorative wreath 810 without the need for any wires and without the need to be proximate to a wired power source. The flexibility offered by the wirelessly powered decorative system 800 is very beneficial to the user, as it enables freedom of placement with respect to the wirelessly powered decorative devices and the decorative objects on which they are deployed. For example, decorative wreath 810 can be placed on a door without any wires and be proximate wireless power transmitter 805, which can be conveniently connected to mains power for the structure that is typically located in the vicinity of a door to a structure.

[0080] As shown in FIG. 8, the wirelessly powered decorative system 800 can include numerous wirelessly powered decorative devices, such as wirelessly powered decorative devices 600G and 600H, that can be deployed on decorative wreath 810. Those of skill in the art will appreciate that decorative wreath 810 can be an artificial wreath, such as one made from faux evergreen foliage, or a wreath made from natural materials, such as fresh cut evergreens. Furthermore, those of skill in the art will appreciate that wirelessly powered decorative system 800 could be used with many decorative objects and applications, such as garland, holiday

swag, decorative foliage, decorative flowers, decorative berries, tree-toppers, decorative Christmas baubles, decorative Easter eggs, and decorations for many other types of holidays, parties, and events. Those of skill in the art will appreciate that placing decorative objects, such as conventional light strings, on such decorations may be problematic and may detract from the decorative aesthetic of the decoration. For example, a wreath made from fresh cut evergreens may be difficult to wrap with a light string. Furthermore, a wrapped light string on a wreath may have many lights that are not visible once the wreath is positioned for display. In some embodiments of the present disclosure, the wirelessly powered decorative devices of the wirelessly powered decorative system **800** can be easily individually configured on decorative wreath **810** for optimum display and decoration without any cumbersome or aesthetically pleasing wires. Those of skill in the art will appreciate that a single wirelessly powered decorative system **800** can be deployed on multiple different decorative objects, such as being deployed on a wreath during Christmas season and deployed on an Easter egg basket during Easter season.

[0081] In some embodiments of the present disclosure, the wirelessly powered decorative devices **600G** and **600H** can include loads such as LED devices. The LED devices in some embodiments of the present disclosure can be illuminated via the wirelessly powered decorative system **800** to provide various colors of light or patterns of light. Those of skill in the art will understand that the LED devices of the wirelessly powered decorative devices **600G** and **600H** can be more brilliantly illuminated in comparison to LED devices associated with conventional battery powered decorative systems. As shown in the embodiment in FIG. **8**, the wirelessly powered decorative system **800** can include a wireless power transmitter **805**. In the embodiment depicted in FIG. **8**, the wireless power transmitter **805** is a small box shaped device. In other embodiments, the wireless power transmitter **805** could be a ring-shaped transmitter device equivalent in diameter to the diameter of the decorative wreath **810**. Those of skill in the art will appreciate that wireless power transmitter **805** can be configured for near field or non-radiative wireless transmission, which enables power to be transferred over short distances by magnetic fields using inductive coupling between coils of wire, or by electric fields using capacitive coupling between metal electrodes. In the embodiment shown in FIG. **8**, the wireless power transmitter **805** can be configured near a power source that is proximate to the decorative wreath **810**. The wireless power transmitter **805** can transmit power to each of the wirelessly powered decorative devices on the decorative wreath **810**, such as wirelessly powered decorative devices **600G** and **600H**. In some embodiments of the present disclosure, the wireless power transmitter **805** can have an unobstructed transmission path between the wireless power transmitter **805** and the wirelessly powered decorative devices, as depicted in FIG. **8**. In other embodiments, the wireless power transmitter **805** can transmit signals via a partially obstructed or fully obstructed path. For example, the wireless power transmitter **805** depicted in FIG. **8** could be positioned on the back side of door **815** and configured to transmit wireless power through the door **815** to the wirelessly powered decorative devices **600G** and **600H** on the other side of door **815**. In another example, the wireless power transmitter **805** depicted in FIG. **8** could be positioned in a discrete and less visible location, such as behind a couch

or other furniture or on book case or other structure, and configured to transmit wireless power via a partially obstructed path to the wirelessly powered decorative devices **600G** and **600H**.

[0082] In some embodiments of the present disclosure, each of the wirelessly powered decorative devices **600G** and **600H** of the wirelessly powered decorative system **800** can provide a receiver to receive the wireless power transmission from the wireless power transmitter **805**. In some embodiments, the receivers are configured to receive the electromagnetic field generated by the wireless power transmitter **805**, and then generate corresponding electrical currents accordingly for consumption by a load, such as an LED device, or to be stored by a battery. Thus, the receivers can provide the generated currents to one or more loads connected to the wirelessly powered decorative devices **600G** and **600H** such as LED lights, ornaments, or tree toppers, such as stars or angels. Those of skill in the art will appreciate that wireless power transmitter **805** can transmit both power and data signals to the wirelessly powered decorative devices. Accordingly, in some embodiments, the wirelessly powered decorative devices include a microcontroller device, which can be configured to receive and process wireless data transmission received by the receiver of the wirelessly powered decorative devices. In these embodiments, the microcontroller of the wirelessly powered decorative devices can be configured to control the operation and display of the wirelessly powered decorative devices.

[0083] Reference is made to FIG. **9**, which is a diagram illustrating a wirelessly powered decorative system **900**, according to some embodiments of the present disclosure. As shown in the embodiment in FIG. **9**, the wirelessly powered decorative system **900** can be implemented on decorative garland **910**. Those of skill in the art will appreciate that decorative garland **910** can be an artificial wreath, such as one made from faux evergreen foliage, or a wreath made from natural materials, such as fresh cut evergreens. The embodiment of the wirelessly powered decorative system **900** shown in FIG. **9** enables the decorative garland **910** to be placed on mantle **915** without the need for a wired connection to a power source or large battery-based power source on the garland **810**. As those of skill in the art will appreciate, mantles, like mantle **915**, are not typically configured with an integrated power source; thus, configuring a decorative system on decorative garland **910** mounted on mantle **915** can be problematic because it may interfere with the operation of the fireplace and/or require wiring that is not aesthetically pleasing and may detract from the desired decorative look. Additionally, conventional battery-based power sources for decorative objects, like decorative garland **910**, are known by those of skill in the art to have numerous drawbacks as such conventional battery-based power sources for decorative systems can be large and cumbersome, detract from the decorative aesthetic, and the batteries must be replaced frequently and often corrode. In accordance with embodiments of the present disclosure, wirelessly powered decorative system **900** be configured to wirelessly provide power to wirelessly powered decorative devices deployed on decorative garland **910** without the need for any wires and without the need to be proximate to the power source. The flexibility offered by the wirelessly powered decorative system **900** is very beneficial to the user, as it enables freedom of placement with respect to the wirelessly powered decorative devices and the items on

which they are deployed. For example, decorative garland **910** can be freely arranged on mantle **915** without any wires and be proximate wireless power transmitter **905**, which can be conveniently connected to mains power for the structure that is typically located in the vicinity of a mantle.

[0084] In some embodiments of the present disclosure, the wirelessly powered decorative devices **6001**, **600J**, and **600K** can include loads such as LED devices. The LED devices in some embodiments of the present disclosure can be illuminated via the wirelessly powered decorative system **900** to provide various colors of light or patterns of light. Those of skill the art will understand that the LED devices of the wirelessly powered decorative devices **6001**, **600J**, and **600K** can be more brilliantly illuminated in comparison to LED devices associated with conventional battery powered decorative systems. As shown in the embodiment in FIG. 9, the wirelessly powered decorative system **900** can include a wireless power transmitter **905**. In the embodiment depicted in FIG. 9, the wireless power transmitter **905** is a small box shaped device. In other embodiments, the wireless power transmitter **905** could be a ring-shaped transmitter device. Those of skill in the art will appreciate that wireless power transmitter **905** can be configured for near field or non-radiative wireless transmission, which enables power to be transferred over short distances by magnetic fields using inductive coupling between coils of wire, or by electric fields using capacitive coupling between metal electrodes. Alternatively, wireless power transmitter **705** can be configured for radio frequency power transfer, such that energy is broadcast in a radio frequency signal. Those of skill in the art will appreciate that wireless transmission of energy, as disclosed herein, can be achieved using an inductive power transfer, capacitive power transfer, laser power transfer, or radio frequency power transfer. In the embodiment shown in FIG. 9, the wireless power transmitter **905** can be configured near a power source that is proximate to the decorative garland **910**. The wireless power transmitter **905** can transmit power to each of the wirelessly powered decorative devices on the decorative garland **910**, such as wirelessly powered decorative devices **6001**, **600J**, and **600K**. In some embodiments of the present disclosure, the wireless power transmitter **905** can have an unobstructed transmission path between the wireless power transmitter **905** wirelessly powered decorative devices, as depicted in FIG. 9. In other embodiments, the wireless power transmitter **905** can transmit signals via a partially obstructed or fully obstructed path. For example, the wireless power transmitter **905** depicted in FIG. 9 could be positioned on the back side of the wall on which the mantle **915** is mounted and configured to transmit wireless power through the wall to the wirelessly powered decorative devices **6001**, **600J**, and **600K**. In another example, the wireless power transmitter **905** depicted in FIG. 9 could be positioned in a discrete and less visible location, such as behind a couch or other furniture or on bookcase or other structure, and configured to transmit wireless power via a partially obstructed path to the wirelessly powered decorative devices **6001**, **600J**, and **600K**.

[0085] The embodiments may further be described using the following clauses:

[0086] 1. An artificial tree comprising: a trunk comprising one or more trunk sections, the one or more trunk sections comprising: a first trunk section comprising: a first transmitter connected to a power source and configured to convert an input power received from the

power source into an electromagnetic field; and a first receiver arranged proximate the first transmitter and configured to receive the electromagnetic field generated by the first transmitter and generate a first output current.

[0087] 2. The artificial tree of clause 1, further comprising: a second trunk section mechanically connected to the first trunk section, the second trunk section comprising: a second receiver configured to generate a second output current in response to the electromagnetic field transmitted between the first trunk section and the second trunk section.

[0088] 3. The artificial tree of clause 1, further comprising: one or more branch portions comprising one or more loads, wherein each of the one or more branch portions is connected to one of the one or more trunk sections, and the one or more loads are powered by one or more corresponding output currents from the one or more trunk sections.

[0089] 4. The artificial tree of clause 1, further comprising: a lighting module configured to receive the first output current,

[0090] 5. The artificial tree of clause 1, wherein the one or more trunk sections further comprise: a second trunk section mechanically connected to the first trunk section, the second trunk section comprising: a second receiver configured to generate a second output current in response to the electromagnetic field transmitted between the first trunk section and the second trunk section; and a third trunk section mechanically connected to the second trunk section, the third trunk section comprising: a third receiver configured to generate a third output current in response to the electromagnetic field transmitted from the first trunk section, via the second trunk section, to the third trunk section.

[0091] 6. The artificial tree of clause 5, wherein the second trunk section further comprises: one or more battery components coupled to the second receiver and configured to be charged using the second output current; and a second transmitter configured to convert power stored in the one or more battery components into the electromagnetic field.

[0092] 7. The artificial tree of clause 6, wherein the second trunk section further comprises: a rectifier configured to rectify an induced alternating current, generated in response to the electromagnetic field, to output the second output current.

[0093] 8. The artificial tree of clause 5, wherein the second trunk section further comprises: one or more intermediate resonators configured to enhance a transfer distance of wireless power transmitted within the second trunk section.

[0094] 9. The artificial tree of clause 1, further comprising: an ornament comprising: a receiver configured to generate an output current in response to the electromagnetic field generated by the first transmitter; a lighting module comprising one or more LED components powered by the output current; and a control unit electrically connected to the one or more LED components and configured to output one or more driving signal to drive the one or more LED components.

[0095] 10. The artificial tree of clause 9, wherein the ornament is a LED light source, a snow globe, a tree topper, a decorative LED candle, a projector, a levitat-

ing object, a decorative candy cane, an imbedded light source in foliage or branches of the artificial tree, or any combination thereof.

- [0096] 11. The artificial tree of clause 1, wherein the first receiver comprises one or more openings, passage-ways, or channels for transmission of the electromagnetic field within the trunk.
- [0097] 12. The artificial tree of clause 1, wherein the first transmitter comprises one or more transmitting coils, and the first receiver comprises one or more receiving coils associated with the one or more transmitting coils.
- [0098] 13. The artificial tree of clause 12, wherein the one or more transmitting coils and the one or more receiving coils are flat spiral coils.
- [0099] 14. The artificial tree of clause 12, wherein the first transmitter and the first receiver communicate wirelessly through inductive coupling via induced magnetic fields between the one or more transmitting coils and the one or more receiving coils.
- [0100] 15. The artificial tree of clause 12, wherein the one or more transmitting coils and the one or more receiving coils are disposed at a core part of the one or more trunk sections, with openings between the core part and an outside rim of a corresponding receiver.
- [0101] 16. The artificial tree of clause 12, wherein the one or more transmitting coils and the one or more receiving coils are disposed contiguous to an outside rim of a corresponding receiver.
- [0102] 17. The artificial tree of clause 16, wherein an open passageway is provided in the core of the one or more trunk sections.
- [0103] 18. The artificial tree of clause 1, wherein the first receiver comprises one or more output connectors configured to be connected to one or more loads of one or more branch portions.
- [0104] 19. The artificial tree of clause 18, wherein the one or more trunk sections are drilled and include one or more holes for mounting the one or more output connectors.
- [0105] 20. An artificial tree comprising: a trunk comprising one or more trunk sections, the one or more trunk sections comprising: a receiver configured to generate an output current in response to an electromagnetic field transmitted to a corresponding trunk section from an external power source; and a lighting module coupled to the trunk, the lighting module comprising one or more LED components powered by the output current.
- [0106] 21. An artificial tree comprising: a trunk comprising one or more trunk sections, the one or more trunk sections comprising: a receiver configured to generate an output current in response to an electromagnetic field transmitted to a corresponding trunk section from an external power source.

[0107] The foregoing outlines features of several embodiments of the present disclosure so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not

depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

- 1. A wirelessly powered decorative system comprising: a decorative item; a wirelessly powered decorative device in communication with the decorative item, the wirelessly powered decorative device comprising a receiver and a load; a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device; and a microcontroller, wherein the load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.
- 2. The wirelessly powered decorative system of claim 1, wherein the load is an LED device and the LED device can be illuminated by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.
- 3. The wirelessly powered decorative system of claim 1, wherein the microcontroller is configured to receive and process wireless data.
- 4. The wirelessly powered decorative system of claim 1, wherein the wireless power transmitter comprises one or more transmitting coils, and the receiver comprises one or more receiving coils associated with the one or more transmitting coils.
- 5. A wirelessly powered decorative system item comprising: a garland strand; a wirelessly powered decorative device in communication with the garland strand, the wirelessly powered decorative device comprising a receiver and a load; a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device; and wherein the load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.
- 6. The wirelessly powered decorative system of claim 5, wherein the load is an LED device and the LED device can be illuminated by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.
- 7. The wirelessly powered decorative system of claim 6, wherein the wirelessly powered decorative device further comprises a microcontroller.
- 8. The wirelessly powered decorative system of claim 5, wherein the wirelessly powered decorative device receives a power signal and a data signal from the wireless power transmitter.
- 9. The wirelessly powered decorative system of claim 5, wherein the wireless power transmitter comprises one or more transmitting coils, and the receiver comprises one or more receiving coils associated with the one or more transmitting coils.
- 10. A decorative object, comprising: a garland strand; a wirelessly powered decorative device in communication with the garland strand, the wirelessly powered decorative device comprising a receiver and a load;

a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device; and

wherein the load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.

11. The decorative object of claim **10**, wherein the decorative object is a wreath, garland strand, artificial tree, holiday swag, or decorative foliage.

12. The decorative object of claim **10**, wherein the receiver comprises one or more receiving coils associated with the one or more transmitting coils of the wireless power transmitter.

13. The decorative object of claim **10**, wherein the decorative object is not connected to by a wire to a power source.

14. The decorative object of claim **10**, wherein the wirelessly powered decorative device further comprises a microcontroller.

15. An artificial tree comprising:

a decorative item;

a wirelessly powered decorative device in communication with the decorative item, the wirelessly powered decorative device comprising a receiver and a load;

a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device; and

wherein the load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.

16. The artificial tree of claim **15**, wherein the load is an LED device and the LED device can be illuminated by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.

17. The artificial tree of claim **15**, wherein the wirelessly powered decorative device further comprises a microcontroller.

18. The artificial tree of claim **15**, wherein the wirelessly powered decorative device receives a power signal and a data signal from the wireless power transmitter.

19. The artificial tree of claim **15**, wherein the wireless power transmitter comprises one or more transmitting coils, and the receiver comprises one or more receiving coils associated with the one or more transmitting coils.

20. An artificial tree comprising:

an ornament;

a wirelessly powered decorative device in communication with the ornament, the wirelessly powered decorative device comprising a receiver and a load;

a wireless power transmitter configured to transmit power to the receiver of the wirelessly powered decorative device; and

wherein the load can be powered by power received by the receiver of the wirelessly powered decorative device from the wireless power transmitter.

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